

Exercise: section 11

Sub sequences

11.1) Let $a_n = 3 + 2(-1)^n \quad \forall n \in \mathbb{N}$

(a) List 1st 8 terms of the seq

$$(1, 5, 1, 5, 1, 5, 1, 5)$$

(b) Given a subseqⁿ that is constant.
Specify the selection function σ

$$\sigma(k) = 2k \quad \lim S_{2k} = 5$$

$$\sigma(k) = 2k+1 \quad \lim S_{2k+1} = 1$$

11.2) Consider the seqⁿ defined as follows

$$a_n = (-1)^n, \quad b_n = \frac{1}{n}, \quad c_n = n^2$$

$$d_n = \frac{6n+4}{7n-3}$$

(a) $\forall \text{ seq}^{\mathbb{N}}$, Give an example of monotone $\text{Subseq}^{\mathbb{N}}$.

$$a_n = (-1)^n \quad (a_{2k})_{k \in \mathbb{N}} = 1$$

$$b_n = \frac{1}{n} \quad (b_{2k})_{k \in \mathbb{N}} = \frac{1}{2k}$$

$$c_n = n^2 \quad (c_{2k})_{k \in \mathbb{N}} = 4k^2$$

$$d_n = \frac{6n+4}{7n-3} \quad (d_{2k})_{k \in \mathbb{N}} = \frac{12k+4}{14k-3}$$

(b) $\forall \text{ seq}^{\mathbb{N}}$, give its set of $\text{Subseq}^{\mathbb{N}}$ limits.

$$a_n = (-1)^n \quad a = \{-1, 1\}$$

$$b_n = \frac{1}{n} \quad b = \{0\}$$

$$c_n = n^2 \quad c = \{+\infty\}$$

$$d_n = \frac{6n+4}{7n-3} \quad d = \left\{ \frac{6}{7} \right\}$$

③ $\forall \text{ seq}^n$, give its $\limsup$, $\liminf$

$$\limsup_{n \rightarrow \infty} a_n = 1 \quad \liminf_{n \rightarrow \infty} a_n = -1$$

$$\limsup_{n \rightarrow \infty} b_n = \liminf_{n \rightarrow \infty} b_n = 0$$

$$\limsup_{n \rightarrow \infty} c_n = \liminf_{n \rightarrow \infty} c_n = +\infty$$

$$\limsup_{n \rightarrow \infty} d_n = \liminf_{n \rightarrow \infty} d_n = \frac{6}{7}$$

④ which of the seq^n converges?
diverges to $+\infty$? diverges to $-\infty$?

b_n converges

c_n diverges to $+\infty$

d_n converges to $\frac{6}{7}$

⑤ which of the seq^n is bounded?

a_n, b_n, d_n are bounded

11.3

repeat Exercise 11.2 for the seqⁿ

$$* S_n = \cos\left(\frac{n\pi}{3}\right)$$

$$* S = \left\{ -1, -\frac{1}{2}, \frac{1}{2}, 1 \right\}$$

$$* \limsup_{n \rightarrow \infty} S_n = 1 \quad \liminf_{n \rightarrow \infty} S_n = -1$$

* Bounded

$$t_n = \frac{3}{4n+1}$$

$$* \lim_{n \rightarrow \infty} t_n = 0 \quad t = \{0\}$$

$$* \liminf_{n \rightarrow \infty} t_n = \limsup_{n \rightarrow \infty} t_n = 0$$

* Converges to 0

* Bounded

$$U_n = \left(-\frac{1}{2}\right)^n$$

$$\uparrow \left(U_{2k} \right)_{k \in \mathbb{N}} = \left(\left(-\frac{1}{2}\right)^{2k} \right)_{k \in \mathbb{N}} = \left(\left(\frac{1}{2}\right)^k \right)_{k \in \mathbb{N}}$$

monotone subseqⁿ

$$\uparrow U = \{ 0 \}$$

$$\uparrow \limsup_{n \rightarrow \infty} U_n = \liminf_{n \rightarrow \infty} U_n = 0$$

$\uparrow$ Converges to 0

$\uparrow$ Bounded

$$U_n = (-1)^n + \frac{1}{n}$$

$$\left(U_{2k} \right)_{k \in \mathbb{N}} = \left(1 + \frac{1}{2k} \right)_{k \in \mathbb{N}}$$

$$\left(U_{2k+1} \right)_{k \in \mathbb{N}} = \left(-1 + \frac{1}{2k+1} \right)_{k \in \mathbb{N}}$$

$\uparrow$ monotonic subseqⁿ

$$\ast \quad V = [-1, 1]$$

$$\ast \quad \limsup_{n \rightarrow \infty} V_n = 1 \quad \liminf_{n \rightarrow \infty} V_n = -1$$

$\ast$ Bounded

(11.4) Repeat Exercise 11.2 for the seqⁿ

$$\textcircled{i} \quad w_n = (-2)^n$$

$$(w_{2k})_{k \in \mathbb{N}} = (4, 16, 64, \dots)$$

monotone increasing subseqⁿ

$$(w_{2k+1})_{k \in \mathbb{N}_0} = (-2, -8, -32, \dots)$$

monotone decreasing subseqⁿ

Subsequential Limit

$$S = \{-\infty, +\infty\}$$

$$\limsup w_n = +\infty$$

$$\liminf w_n = -\infty$$

Both of them are unbounded.

(ii) $x_n = 5(-1)^n$

$$(x_{2k+1})_{k \in \mathbb{N}_0} = \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \dots \right)$$

$$(x_{2k})_{k \in \mathbb{N}} = (5, 5, 5, \dots)$$

Both are monotone, Bounded

$$S = \left[\frac{1}{5}, 5 \right]$$

$$\limsup x_n = 5$$

$$\liminf x_n = \frac{1}{5}$$

$$(iii) \quad y_n = 1 + (-1)^n$$

$$(y_{2k+1})_{k \in \mathbb{N}_0} = (0, 0, 0, \dots)$$

$$(y_{2k})_{k \in \mathbb{N}} = (2, 2, 2, \dots)$$

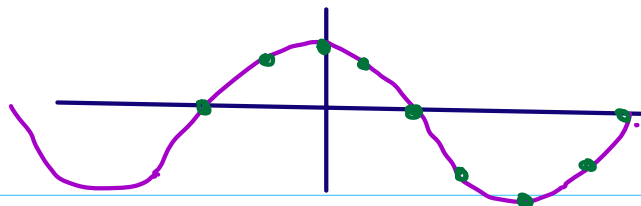
Both of them are monotone
and bounded.

$$S = [0, 2]$$

$$\limsup y_n = 2$$

$$\liminf y_n = 0$$

iv



$$z_n = n \cos\left(\frac{n\pi}{4}\right)$$

$$\begin{aligned} (z_{8k})_{k \in \mathbb{N}_0} &= \left(8k \cos\left(\frac{8k\pi}{4}\right) \right) \\ &= (8k) \\ &= (0, 8, 16, 24, \dots) \end{aligned}$$

$$\begin{aligned} (z_{8k+1})_{k \in \mathbb{N}_0} &= \left((8k+1) \cos\left(\frac{(8k+1)\pi}{4}\right) \right) \\ &= \left((8k+1) \cdot \frac{1}{\sqrt{2}} \right)_{k \in \mathbb{N}_0} \\ &= \left(\frac{1}{\sqrt{2}}, \frac{9}{\sqrt{2}}, \frac{17}{\sqrt{2}}, \dots \right) \end{aligned}$$

$$\begin{aligned} (z_{8k+2})_{k \in \mathbb{N}_0} &= \left((8k+2) \cos\left(\frac{(8k+2)\pi}{4}\right) \right) \\ &= \left((8k+2) \cdot 0 \right) \end{aligned}$$

$$= (0, 0, 0, \dots)$$

$$(z_{8k+3})_{k \in \mathbb{N}_0}$$

$$= \left((8k+3) \cdot \cos\left(\frac{(8k+3)\pi}{4}\right) \right)_{k \in \mathbb{N}_0}$$

$$= \left((8k+3) \cdot -\frac{1}{\sqrt{2}} \right)$$

$$= \left(-\frac{3}{\sqrt{2}}, -\frac{11}{\sqrt{2}}, -\frac{19}{\sqrt{2}}, \dots \right)$$

$$(z_{8k+4})_{k \in \mathbb{N}_0}$$

$$= \left((8k+4) \cdot -1 \right)_{k \in \mathbb{N}_0}$$

$$= (-4, -12, -20, \dots)$$

Subsequential Limit Points

$$S = \{-\infty, 0, +\infty\}$$

$$\limsup z_n = +\infty$$

$$\liminf z_n = -\infty$$

11.5 Let (q_n) be an enumeration of all the rational's in the interval $(0, 1]$

(a) Give the set of subsequential limit's for (q_n)

Solⁿ

Claim: Every $t \in [0,1]$ is a subsequential limit iff for every $\varepsilon > 0$, the set $\{n \in \mathbb{N} : |a_n - t| < \varepsilon\}$ is infinite

So for fixed $t \in [0,1]$, and

$\varepsilon > 0$, we have to show

there are infinitely many a_n in the $(t - \varepsilon, t + \varepsilon)$

But by the definition of denseness of

$\mathbb{Q}$, for every $c < d \in \mathbb{Q} \cap [0,1]$

$\exists x \in \mathbb{Q}$ such that $c < x < d$

let's construct a seqⁿ where

$$(1) \quad t - \varepsilon < x_1 < t + \varepsilon$$

$$(2) \quad \text{if } x_1 < t, \quad x_1 < x_2 < t + \varepsilon$$

$\vdots$

$\Rightarrow$ There are infinitely many rational

$$\text{in } (t - \varepsilon, t + \varepsilon) \quad \forall \varepsilon > 0 \Rightarrow t$$

is a subsequential limit.

$\Rightarrow$ for $q_n \in [0, 1]$ the subsequ

$$\text{limit point's set} = [0, 1]$$

$$\limsup q_n = 1$$

$$\liminf q_n = 0$$

11.7) let (a_n) be an enumeration of the set $\mathbb{Q}$ of all rational numbers. Show there exists a subsequence (a_{n_k}) such that

$$\lim_{k \rightarrow \infty} a_{n_k} = +\infty$$

take subsequence such that

$$a_{n_k} = \frac{n_k}{1} \in \mathbb{Q}$$

$$\lim_{n \rightarrow \infty} n = +\infty$$

11.8) Use Definition 10.6 and Exercise 5.4 to prove $\liminf s_n = -\limsup (-s_n)$ for every sequence (s_n)

Solⁿ

Definition 10.6:

$$\limsup s_n = \lim_{N \rightarrow \infty} \sup \{ s_n : n > N \}$$

$$\liminf s_n = \lim_{N \rightarrow \infty} \inf \{ s_n : n > N \}$$

Exercise 5.4:

$$-S = \{ -s : s \in S \}.$$

$$\Rightarrow \inf S = -\sup(-S)$$

$$-\limsup(-s_n)$$

$$= -\lim_{N \rightarrow \infty} \sup \{ -s_n : n > N \}$$

$$= \lim_{N \rightarrow \infty} -\sup \{ -s_n : n > N \}$$

$$= \lim_{N \rightarrow \infty} \inf \{ s_n : n > N \}$$

$$= \liminf S_n$$